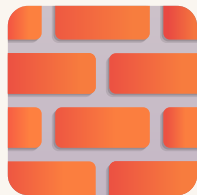


WEEK 7



Pistons and Mazes

Power a bridge, then cross

피스톤으로 길을 열고 건너기 (영어코딩)

BIG PICTURE

Machines that move blocks



A sticky piston pulls a block back and opens a path — while
powered.

레버로 피스톤을 켜면 다리가 열려요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a piston?

A piston **pushes** a block when it gets a redstone signal. A **sticky** piston can also **pull** it back.

```
lever ON → piston pulls block → path opens  
lever OFF → piston pushes back → path closes
```

신호가 있으면 열리고, 끊기면 닫혀요

CONCEPT

Only flip when there's a gap

```
if agent.detect(BLOCK, FORWARD):  
    agent.interact(FORWARD)    # power the piston  
else:  
    agent.move(FORWARD, 1)
```

Interact only at the blocked spot — not on every step.

막힌 곳에서만 레버를 작동시켜요

CODE

Power, then cross

```
agent.move(FORWARD, 1)
agent.interact(FORWARD)    # piston opens the gap
agent.move(FORWARD, 2)    # cross while it's open
```

Cross **before** the signal turns off and the bridge vanishes.

다리가 열려 있는 동안 건너요

WATCH IT!

Signal opens the bridge

`agent.interact(FORWARD)`



redstone signal → piston



sticky piston pulls block



gap opens → cross ✓

CODE

Loop across the maze

```
while not agent.detect(BLOCK, DOWN):  
    if agent.detect(BLOCK, FORWARD):  
        agent.interact(FORWARD)  
        agent.move(FORWARD, 1)  
    else:  
        agent.move(FORWARD, 1)
```

At a gap → power + cross. On solid ground → just walk.

PREDICT 🤖

Can the agent move now?

Predict first — then click reveal

```
agent.move(FORWARD, 1)
agent.interact(FORWARD)    # sticky piston pulls
agent.move(FORWARD, 1)
```

Reveal output 🔍 reveal

VOCAB

New words today

piston

a block that pushes others when powered

피스톤

sticky piston

a piston that can pull a block back too

끈끈이 피스톤

gap

a missing section of the path

틈

bridge

the block the piston moves into place

다리



DEBUG

What's wrong with this code?

Goal: interact only when a gap is ahead.

buggy

```
while not agent.detect(BLOCK, DOWN):  
    agent.interact(FORWARD)  
    agent.move(FORWARD, 1)
```

Think first — then click next.



DEBUG

The bug

Bug: it interacts on *every* step, flipping levers it shouldn't. Wrap `interact` in an `if` so it only fires when a block is ahead.

fixed

```
while not agent.detect(BLOCK, DOWN):  
    if agent.detect(BLOCK, FORWARD):  
        agent.interact(FORWARD)  
    agent.move(FORWARD, 1)
```

YOUR TURN!

Cross the gap

- ✓ Open the Week 7 world + Code Builder
- ✓ Move to the lever and power the piston
- ✓ Cross *while* the path is open
- ✓ Only **interact** when a gap is ahead
- ✓ Send your walkthrough video on KakaoTalk



QUIZ

QUESTION 1 OF 3

What makes a piston move a block?

A redstone signal, like one from a flipped lever.

The agent standing nearby.

Nothing — pistons never move.

Daylight.



QUIZ

QUESTION 2 OF 3

What can a **sticky** piston do that a normal one can't?

Nothing different.

Fly.

Move two blocks at once.

Pull the block back when the signal stops.



QUIZ

QUESTION 3 OF 3

Why guard `interact` with `if agent.detect(BLOCK, FORWARD)` ?

So it only flips a lever when something is actually ahead.

It has no effect.

So it moves faster.

So the loop never ends.

QUIZ

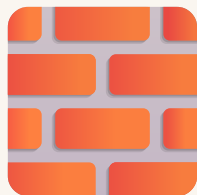
Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Bridge crossed!

Next: the $20 \times 20 \times 20$ capstone maze.

다음 주: 최종 $20 \times 20 \times 20$ 큐브 미로